Assignment-Dependent Charging and Marginal-Price Support in Electric-Bus Fleets

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Abstract

We characterize when a complete electric-bus fleet minimizes its operating cost and electricity bill at the marginal supply prices associated with its own load. For heterogeneous services in two simultaneous batches, a declared successor graph and fixed shared charging resources, we derive the exact joint load–vehicle-cost frontier with full battery replenishment. Cardinality-matching thresholds determine where a convexified optimum has a physical implementation. The load-only set is an interval; its discrete vehicle-cost layers cause the obstruction. A bidirectional extension accounts for gross connector throughput and signed electricity settlement. Vehicle-to-grid operation can preserve or remove a positive gap, and identical paid idle buses are redundant under a stated fleet cap. Exact examples and parameter regions distinguish useful flexibility from mere enlargement of the duty set. The matching and price-support machinery is classical; the contribution is the complete physical classification under these explicit assumptions. For one public 37-service timetable, an exact feasible response proves that the best certified incumbent is unsupported, while global optimal-fleet support remains unresolved. Synthetic coefficients and ideal common-window physics limit conclusions about operational prevalence or calibrated value.

1 Introduction

An electric-bus operator assigns mandatory services to vehicles and decides when to recharge them. Continuous charging offers flexibility, but a vehicle cannot cover a fraction of a service. Assigning two services to the same bus can therefore impose an intermediate energy requirement that disappears when two buses are used. A tariff may reward the resulting change in charging without compensating the operator for its additional vehicle.

We ask a precise question. A supplier has a convex incremental cost for serving aggregate fleet demand. A tariff is calculated from a candidate fleet's load and posted before the operator chooses its complete schedule. Holding that tariff fixed, does the candidate minimize the operator's bill plus its intrinsic operating cost? This is an own-load price-support test. Its participant is one operator controlling the complete fleet. It is not a strategic model in which that operator anticipates a new tariff after each alternative it considers.

The connection between price support and convexification is established in nonconvex market theory. We develop its physical implications for a class of heterogeneous services in two simultaneous batches. Every assignment allowed by the declared successor graph is retained, with one fixed shared connector in each charging window. An exact joint energy–vehicle-cost projection reduces the complete fleet set to early-energy thresholds indexed by matching cardinality; standard matching convexity then identifies the hull and a precise support criterion. The energy-only feasible set is an interval; the relevant nonconvexity concerns its cost-labelled assignment layers. Full terminal

replenishment is central: an initially full battery must not provide free net energy over a finite horizon. Every plan buys the same total service energy, yet its assignment changes *when* that energy must be purchased, coupled with the fixed cost of a used bus.

Bidirectional charging changes this frontier through both available battery inventory and connector time. Charging one bus while discharging another does not cancel the time spent on a single connector: its resource is gross throughput. We derive the resulting signed-load thresholds and show exactly how the obstruction can persist or disappear. Under a total fleet cap, identical paid idle buses add no projected alternatives beyond those obtained by separating matched service pairs. This statement concerns the declared ideal model, not the value of spare buses in general V2G operations.

The heterogeneous class makes the assignment set exhaustive under explicit timing and resource assumptions. A two-service specialization provides an accessible illustration and an exact connector-capacity and vehicle-fee characterization. We then distinguish two kinds of enlargement. Additional energy-consuming movements under unchanged equipment and charging access cannot improve either the physical or hull value in the stated construction. An added battery type changes the equipment set and can restore support. Its fee matters: a newly admitted plan cheaper than the old hull optimum need not restore support because the enlarged hull can improve further. Finally, an exact replication calculation shows why a small planning gap and a small whole-owner incentive are different statements.

These are application-specific results in a synthetic model. We do not claim that nonconvex demand or convex-hull price support is a new concept. The public timetable evidence is included to state clearly what is known at a larger scale and what remains unproved. Its bounds do not currently show that its optimal physical fleet is unsupported. Practical magnitude and prevalence are separate research questions.

2 Relation to existing work

Electric-bus scheduling combines service assignment, battery feasibility and charging access. Grid-aware multi-depot scheduling, capacitated stations and scalable route-and-recharge generation are studied by Wu et al. [18], de Vos et al. [5], Parmentier et al. [13]. These operational problems motivate our use of a complete fleet: charging feasibility is joint across the selected vehicle duties, especially at a shared connector. Our question concerns support over alternatives that can change both the number of used buses and the charging opportunities created by their duties. Integrated flow and set-partitioning models already combine assignments, battery feasibility and charging slots [4]. Wu et al. [18] already choose vehicle-to-trip assignments and charging under grid-related objectives; full overnight charging is among their assumptions. Endogenous assignments, full charging and grid-aware scheduling are therefore not, separately or together, sufficient novelty claims here.

Zoltowska and Lin [21] formulate aggregation and disaggregation of bus charging around hourly auction bids using integer models. Their work is a precedent for discrete bus charging and market participation; our question additionally allows service assignments and used-bus costs to change under explicit terminal replacement-energy accounting. More directly, Wu et al. [19] describe bus service planning coupled to market clearing and distribution locational marginal prices in a bilevel framework. Our question is the support of complete physical fleets at a fixed posted marginal tariff, rather than a first coupling of bus operations to endogenous electricity prices.

Price-support failure with indivisible operating decisions is established in electricity-market research. O'Neill et al. [12] study prices in markets with nonconvexities; Gribik et al. [6] discuss market-clearing prices and uplift. Convex-hull pricing provides the relevant convexification frame-

work [14, 8]. Linear-price support in decentralized problems with integralities is also studied through complete linear descriptions [9]. We use this machinery; we do not present the support equivalence below as a new general theorem. Whole feasible participant schedules as columns and their fixed-price self-scheduling oracles also have established decomposition treatments [2]. The complete-fleet unit ensures the right physical alternatives in this application; it is not new column-generation machinery.

Exact EV flexibility aggregation is also established. Mukhi and Abate [10] characterize fixed-population charging polytopes using permutahedra, with prescribed availability and full departure charge. Mukhi et al. [11] give generalized-polymatroid representations and exact Minkowski aggregation for convex individual flexibility sets. Our distinction is the joint load–vehicle-cost frontier when service pairing and the number of used buses change. The load-only projection in our class is an interval, not a new nonconvex flexibility set. Fixed-matching interval aggregation and ordered matching are classical ingredients.

Bidirectional charging and transport decisions have also been combined previously. Blatiak et al. [3, Sections 2.1–2.2] optimize commercial fleet trip timing and charging against energy and balancing-service prices. The abstract of Zhou et al. [20] describes multi-depot bus charging/discharging, spare-bus revenues and cross-line operation. Our comparison with that work is limited to its abstract; we make no claim about the absence of related structural results in its full formulation. Neither coupling transport to V2G nor using spare buses for grid services is a novelty claim here.

Signed aggregate flexibility is established as well. Al Taha et al. [1, Section II, Example 1] model lossless bidirectional batteries with initial and final energy requirements and aggregate a fixed population through Minkowski sums. Our V2G extension instead classifies the joint load–fleet-cost frontier created by successor matching and a shared gross-throughput budget. Its thresholds and paid-idle elimination argument concern that restricted physical class; the persistence and restoration examples illustrate the classification, rather than establishing a first signed-aggregation or V2G price-support result.

Aggregation must specify who owns the aggregated decisions. The classical perspective of Starr [17] and dispatchable-demand results of Hreinsson et al. [7] explain why aggregate nonconvexities can become small. Our replication calculation instead isolates a distinction between a planning-value gap, one owner's absolute fixed-price incentive and the incentives of independently reserved service pairs. The explicit rates do not contradict approximate aggregation results.

Our structural result concerns the joint physical load–cost projection of heterogeneous two-batch services with declared successor matchings, full inventories and stated charging resources. Standard bipartite-flow integrality and LP value-function convexity enter the proof; general matching and price-support theory are not new claims. The resulting threshold criterion explains precisely where the optimal whole-plan mixture lacks a physical implementation. The two-service parameter and equipment regions then isolate changes that preserve or erase the obstruction. Movement-only stability is narrower: its additional alternatives are dominated under the preserved resources. All these conclusions are conditional on their physical and tariff models.

3 Complete fleets and the meaning of support

Fix mandatory services, directed movement opportunities, a charging horizon and physical equipment. A complete plan $x \in \mathcal{X}$ covers every mandatory service exactly once, assigns the services to used buses, routes those buses, and schedules charging subject to battery and shared-resource constraints. Unused buses contribute neither energy purchases nor operating cost. Let $e(x) \in \mathbb{R}^T_+$

denote purchased grid energy by market interval and $c(x)$ intrinsic operating cost. Each used battery starts full and finishes at the same required full inventory. Assume the projected image

$$S = \{(e(x), c(x)) : x \in \mathcal{X}\}$$

is nonempty and compact. All physical planning, price-response and hull problems below use exactly this common set.

Let $F : \mathbb{R}^T \to \mathbb{R} \cup \{+\infty\}$ be proper, closed and convex, finite on all feasible loads. It represents incremental supply cost conditional on fixed background demand. Define

$$D = \min_{x \in \mathcal{X}} \{c(x) + F(e(x))\}, \tag{1}$$

$$\mathrm{CH} = \min_{(e,c) \in \operatorname{conv} S} \{c + F(e)\}, \qquad \Delta = D - \mathrm{CH} \geq 0. \tag{2}$$

The hull mixes *complete feasible plans*; it is not a duty-column relaxation unless equivalence is separately proved. Its mean load need not be executable as a physical daily schedule. Evaluating supply cost at that mean also differs from averaging the costs of plans operated on different days.

For a posted tariff $p \in \mathbb{R}^T$, the complete-fleet response and regret are

$$V(p) = \min_{y \in \mathcal{X}} \{c(y) + p^\top e(y)\}, \qquad r(x; p) = c(x) + p^\top e(x) - V(p). \tag{3}$$

The posted tariff is fixed across alternatives y. A strategic operator anticipating a tariff reset would face a different objective and needs a separate settlement model.

Definition 1 (Weak own-load support)**.** A physical plan x is supported if there exists an admissible marginal price $p \in \partial F(e(x))$ for which $r(x; p) = 0$. Alternative plans may tie. If an economic convention restricts the price set, existence is tested only in that declared subset.

For differentiable supply we use its gradient. For the nonnegative-domain quadratic below this means the gradient of its unrestricted quadratic formula, including at a zero load. Domain normals are not additional generation-price choices. With nonsmooth supply, a failed support test at one chosen multiplier does not exclude support at another admissible price.

For the following classical equivalence, assume either

$$\operatorname{ri} \operatorname{conv} e(\mathcal{X}) \cap \operatorname{ri} \operatorname{dom} F \neq \varnothing,$$

or that a finite convex minorant on $\mathbb{R}^T$ agrees with F on all hull loads. The latter applies to our quadratic. These assumptions ensure an attained hull-dual price; a restricted economic price set must contain such a price for the corresponding converse.

Proposition 2 (Classical support criterion)**.** *For every physical x and $p \in \partial F(e(x))$,*

$$r(x; p) \geq c(x) + F(e(x)) - \mathrm{CH} \geq \Delta. \tag{4}$$

Under the stated qualification, $D = \mathrm{CH}$ if and only if some physical plan is supported. If $D = \mathrm{CH}$, every physical planner optimum is supported by every admissible *attained maximizer of $V(p) - F^*(p)$, where $F^*(p) = \sup_u \{p^\top u - F(u)\}$.*

This specializes established nonconvex support and convexification arguments [12, 6, 8]; the proof is included in Appendix C.1 to make assumptions and quantifiers explicit. It does not say that every subgradient supports a planner optimum, or that a price-adjustment algorithm converges. Without a price restriction, every attained dual maximizer has zero regret and is a subgradient at each physical planner optimum. An external price restriction adds a separate admissibility test to that mathematical conclusion.

Remark 3 (Admissibility is a separate quantifier). For a single physical point $(e, c) = (0, 0)$ and $F(e) = |e|$, both values are zero and every $p \in [-1, 1]$ attains the hull dual. If the declared admissible price set is $\{0\}$, the plan has admissible support at zero; the dual maximizer $p = 1$ remains mathematically supporting but is inadmissible. Containing one dual maximizer does not mean containing them all.

The participant in (3) controls all buses. Shared infrastructure does not automatically define unilateral alternatives for separately owned vehicles. Section 8 introduces reserved rights before making an individual-participant comparison.

4 Complete assignments and their load–cost frontier

The relevant projection keeps operating cost as well as load. A whole interval of early loads can be physical while the cheapest vehicle count changes discontinuously across it. We characterize this joint frontier under fixed shared resources and the declared allowed successor graph. A separate adequate-resource policy is retained in Appendix A.

There are $n \geq 1$ mandatory depot-to-depot services A_i in $[0, 1]$ and n services B_j in $[2, 3]$. Batteries have capacity $B > 0$ and $0 < a_i, b_j \leq B$. A fixed declared bipartite graph Γ specifies which A-to-B shared-bus assignments are allowed. Every allowed transition has zero movement time, energy and cost. Every service retains its standalone bus option and its original depot charging access. At most $2n$ buses are used, each at fee $f > 0$, starting full and returning full by hour 4. One shared early connector has power $K \geq 0$ in $[1, 2]$; one shared terminal connector has power $H \geq 0$ in $[3, 4]$. Each bus accepts the corresponding connector power. There is no other charging, reserve, loss, taper, auxiliary draw or switching time. Powers K, H are fixed resources, not multiples of n. The graph is an operational compatibility restriction, not a selected bank of duties used to approximate a larger declared opportunity set. Write ν for its maximum matching cardinality, including $\nu = 0$.

Define

$$S = \sum_i a_i + \sum_j b_j, \quad L = \max\{0, S - H\}, \quad U = \min\{\sum_i a_i, K\}, \quad r_{ij} = [a_i + b_j - B]_+,$$

$$r_m = \min_{\substack{M \subseteq \Gamma \\ |M| = m}} \sum_{(i,j) \in M} r_{ij}, \qquad r_0 = 0, \qquad c_m = f(2n - m),$$

where M ranges over allowed partial bipartite matchings and $m = 0, \ldots, \nu$. Every such cardinality exists by deleting edges from a maximum matching. A matching of size m uses $2n - m$ buses.

Theorem 4 (Fixed shared-resource load–cost projection)**.** *The model is feasible exactly when $L \leq U$. In that case put*

$$m_{\max} = \max\{0 \leq m \leq \nu : r_m \leq U\}, \qquad m_0 = \max\{0 \leq m \leq \nu : r_m \leq L\}.$$

Its complete physical projection is

$$\bigcup_{m=0}^{m_{\max}} \{(E, S - E, c_m) : \max\{L, r_m\} \leq E \leq U\}. \tag{5}$$

The thresholds r_m are nondecreasing and discretely convex. The lowest operating-cost boundary $C(E)$ of the complete-plan hull starts at (L, c_{m_0}), visits (r_m, c_m) for $m_0 < m \leq m_{\max}$, and is horizontal at $c_{m_{\max}}$ on $[\max\{L, r_{m_{\max}}\}, U]$.

For a fixed matching, early purchases satisfy $0 \le x_i \le a_i$ and $x_i \ge r_{ij}$ for a paired service. Their sum ranges over $[R(M), \sum_i a_i]$. Serialize them at power K: the shared early constraint is exactly $E \le K$. Every bus is present for terminal charging, and individual refill demands sum to $S - E$. Serialization at power H is possible exactly when $S - E \le H$. Zero-power cases require zero energy and no division. Appendix B proves both directions and the frontier geometry. Cardinalities above ν do not exist; those with $r_m > U$ are resource-infeasible. Neither supplies a fictitious upper-clipped endpoint. The energy-only feasible set is simply $[L, U]$; it is the joint cost frontier that matters for convexification.

Corollary 5 (Support from the cost-labelled frontier)**.** *Suppose $L \le U$ and $G(E) = F(E, S - E)$ is finite, continuous and strictly convex on $[L, U]$. Let E^* uniquely minimize $C(E) + G(E)$. Then*

$$D = \mathrm{CH} \quad \Longleftrightarrow \quad E^* \in \{L\} \cup \{r_m : m_0 < m \le m_{\max}\} \cup [\max\{L, r_{m_{\max}}\}, U]. \tag{6}$$

The gap is positive precisely in the open intervals between adjacent retained thresholds. Admissible marginal-price support additionally requires Proposition 2's supply and price-attainment qualification.

The clipped endpoint L and intermediate physical thresholds on collinear edges must be retained. If $L = U$ or $m_0 = m_{\max}$ the gap is zero. At $f = 0$ the zero-matching branch attains every load at zero operating cost, so the positive-fee interior criterion would fail. The empty matching remains available even with isolated vertices or no allowed edges. Consequently the feasibility test is unchanged; when $\nu = 0$, the entire projection has operating cost $2nf$ and its gap is zero.

Corollary 6 (Classical ordered matching reduction for the complete graph)**.** *If Γ contains every A-to-B edge, then for increasingly sorted energies $a_{(i)}, b_{(i)}$,*

$$r_m = \sum_{i=1}^{m} [a_{(i)} + b_{(m+1-i)} - B]_+.$$

Monotonicity permits selecting the m smallest energies in each batch; a convex positive-part exchange inequality then favors opposite pairing. The appendix proves this classical assignment reduction. No matching algorithm novelty is claimed. The shortcut does not extend to arbitrary allowed graphs. With A energies $(5, 15)$, B energies $(14, 15)$ and battery capacity 20, allowing only the two corresponding-index edges gives $r_2 = 10$; the complete graph's crossed assignment gives $r_2 = 9$. The general theorem uses the graph's actual minimum-cost matching values and never invents a forbidden edge.

The fixed allowed-graph extension has independent source-only proof review; its sparse-graph implementation qualification has not been performed. The complete-crossmatching specialization has a separate, prospectively frozen fourteen-case exact implementation qualification. Raw SOC enumeration and serial charging witnesses matched the sorted-threshold formulas in all fourteen cases; all twenty-four negative controls and the smoke anchor passed. The cases include a physically attained intermediate collinear threshold and a supported clipped lower boundary, as well as zero-power, zero-fee, infeasible and continuous-response cases. This is an internal implementation check, separate from the eight-case adequate-resource qualification in Appendix A. Fixed chargers also impose throughput limits as service demand grows; this is no claim of indefinite scalability with n. Nonzero transition consumption would make total energy matching-dependent; transition time or staggered charging windows would invalidate the common- window serial argument. Inserting an arbitrary directed timetable graph without addressing those changes is therefore outside this theorem.

4.1 A fixed single-connector heterogeneous obstruction

Take $n = 2$, $B = 20$, $f = 7$, A energies $(5, 15)$ and B energies $(14, 15)$, with every crossmatching admitted. Use one early connector of power $K = 15$ and one terminal connector of power $H = 49$. The early-requirement matrix is

$$(r_{ij}) = \begin{pmatrix} 0 & 0 \\ 9 & 10 \end{pmatrix}.$$

The crossed full matching requires 9 early kWh; the corresponding-index full matching requires 10. Thus $r_0 = r_1 = 0$, $r_2 = 9$, $S = 49$, $L = 0$ and $U = 15$. Theorem 4 gives the complete physical projection:

Matched services m	Early energy interval	Operating cost
0	$[0, 15]$	28
1	$[0, 15]$	21
2	$[9, 15]$	14

Use the convex supply

$$F(E, L) = 4E + \frac{E^2 + L^2}{20}, \qquad G(E) = F(E, 49 - E) = \frac{2401}{20} - \frac{9E}{10} + \frac{E^2}{10}.$$

The physical optimum is the crossed two-bus plan at $E = 9$. The hull joins $(0, 21)$ to $(9, 14)$ and minimizes at $E^* = 151/18$, strictly inside that edge. Exactly,

$$D = \frac{2681}{20}, \qquad \mathrm{CH} = \frac{434201}{3240}, \qquad \Delta = \frac{121}{3240} > 0. \tag{7}$$

At the physical optimum's marginal price $(49/10, 4)$, its bill plus operating cost is $2181/10$; the one-pair zero-early complete fleet costs 217, so its full-opportunity-set regret is $11/10$.

Both hull endpoints have serial full-inventory witnesses under these fixed shared resources. At zero early charge, pair the 5-energy A with the 14-energy B and serve the other two services separately. Its three terminal demands 19, 15 and 15 total 49 and serialize on the one 49 kW connector in the terminal hour. At early charge9, use the crossed full matching: the 15-energy A bus charges for 9/15 hour on the one 15 kW early connector. Both buses then require 20 terminal kWh, serializing in 40/49 hour on the terminal connector. Every used bus starts and ends full. Weights $11/162$ and $151/162$ on these two complete plans give the hull point. The mean is a whole-plan convexification, not a physical fractional fleet.

The fixed shared-resource, complete-crossmatching implementation was separately qualified on fourteen prospectively fixed exact cases, including this smoke anchor, collinear thresholds, clipped boundaries, zero powers, fee zero, infeasibility and full response continua. All twenty-four negative controls passed. This is internal qualification, not operational prevalence; it does not qualify sparse allowed graphs or the later V2G extension. Appendix A preserves the different n-parallel- connector policy, its continuous value formulas and its separate eight-case qualification. It has the same joint projection for this example but different individual allocation sets; none of its parallel witness timings is used here.

The main theorem admits fixed shared powers and an arbitrary declared allowed graph, but retains two simultaneous batches and zero-time, zero-energy allowed transitions. It does not cover the public varied-time case. The following two-service model makes the same physical/hull distinction transparent and shows how connector and equipment changes can preserve or remove it.

5 Bidirectional early charging under an explicit settlement rule

The two-batch model also permits an exact extension in which assigned buses can discharge before the second batch. This changes both the physical opportunity set and the tariff interpretation. We assume ideal efficiency one, symmetric charging and discharging acceptance, and the *same* posted price for signed net purchases and exports in a market interval. An asymmetric buy/sell tariff or a lossy battery is a different model.

Retain n A services in $[0,1]$, n B services in $[2,3]$, capacity $B>0$, consumptions $0<a_i,b_j\le B$, and the fixed allowed successor graph Γ of Section 4. Every singleton remains available; allowed transitions have zero time, energy and cost. A matching of size m uses exactly $N_m=2n-m$ assigned buses, each with fee $f>0$, full initial inventory and full terminal replenishment. We first derive the projection with service-assigned buses, then allow identical paid idle buses below. In particular, a B-only bus is assigned to service, starts full and is available at the depot throughout the early window.

All assigned buses can use the single early connector during $[1,2]$ in either direction at power magnitude $K\ge 0$. Its unit-window resource is *gross* throughput: if bus v has signed net early purchase q_v, then $\sum_v |q_v|\le K$ must be realizable. A constraint on $|\sum_v q_v|$ alone would incorrectly allow charging and discharging to cancel connector time. Terminal replenishment in $[3,4]$ uses one charging-only connector of power $H\ge 0$. Each bus accepts those corresponding powers. There is no other charging, reserve, auxiliary draw, taper or switching time.

Net early load $E=\sum_v q_v$ may now be negative. The nonnegative fleet-load convention in Section 3 is therefore replaced explicitly for this section: F must be defined and finite on all admitted signed loads $(E,S-E)$. Alternatively, declare fixed background demand and an incremental cost $F(e)=C(d+e)-C(d)$ whose domain contains the whole feasible signed-load set. Export settlement still uses the same linear rate as purchases. There is no implicit extension of a cost defined only on nonnegative fleet demand. If aggregate export is prohibited, impose $E\ge 0$ explicitly; the lower thresholds below are then additionally clipped at zero.

Write $S=\sum_i a_i+\sum_j b_j$, $P=\sum_i a_i$, and let ν be the maximum matching cardinality of Γ. Retain the matching values

$$r_m=\min_{\substack{M\subseteq\Gamma\\ |M|=m}}\sum_{(i,j)\in M}[a_i+b_j-B]_+,\qquad 0\le m\le\nu,$$

with $r_0=0$. The matching machinery is unchanged; the available assigned battery inventory and gross connector budget create the new thresholds.

Lemma 7 (Minimum gross energy for a fixed matching)**.** *For a fixed allowed matching M of size m, put*

$$R=\sum_{(i,j)\in M}[a_i+b_j-B]_+,\qquad T=S-N_mB.$$

Before the shared connector constraint, signed early charges have total $E\in[T,P]$. At a prescribed such E, their minimum gross throughput is

$$\min_{\sum_v q_v=E}\sum_v|q_v|=\max\{E,2R-E\}. \tag{8}$$

The minimum is attained by one monotone action of the chosen sign per bus.

Theorem 8 (Signed-load projection and joint frontier)**.** *Under the above assumptions, define*

$$t_m=\max\{S-H,\ S-(2n-m)B,\ 2r_m-K\},\qquad U=\min\{P,K\},\qquad c_m=f(2n-m). \tag{9}$$

The complete physical projection is exactly the union of

$$\{(E, S-E, c_m) : t_m \le E \le U\} \tag{10}$$

over $0 \le m \le \nu$ *with* $t_m \le U$. *The model is feasible exactly when* $t_0 \le U$. *The sequence* t_m *is nondecreasing and discretely convex.*

When feasible, let $m_{\max} = \max\{m \le \nu : t_m \le U\}$ *and* $m_0 = \max\{m \le m_{\max} : t_m = t_0\}$. *The hull's lowest operating-cost boundary starts at* (t_0, c_{m_0}), *visits* (t_m, c_m) *for* $m_0 < m \le m_{\max}$, *and is horizontal at* $c_{m_{\max}}$ *on* $[t_{m_{\max}}, U]$. *Physical thresholds on collinear hull edges remain part of this statement.*

Corollary 9 (Support and physical attainment)**.** *Suppose* $t_0 \le U$ *and* $G(E) = F(E, S-E)$ *is finite, continuous and strictly convex on* $[t_0, U]$. *Let* E^* *be the unique minimizer of the hull operating boundary plus* G. *Then, for* $f > 0$,

$$D = \mathrm{CH} \quad \Longleftrightarrow \quad E^* \in \{t_0\} \cup \{t_m : m_0 < m \le m_{\max}\} \cup [t_{m_{\max}}, U].$$

Admissible own-load support additionally requires the framework's convex- supply, dual-attainment and price-set qualification on the declared signed domain. At $f = 0$, *the empty-matching branch has zero operating cost over the entire feasible interval and the gap is zero, so the positive-fee strict-interior criterion must not be applied.*

5.1 Why the projection is exact

Proof of Lemma 7. A paired bus has signed charge in $[a_i + b_j - B, a_i]$: early headroom gives the upper bound, and enough inventory for B gives the lower bound. An A-only bus has $q \in [a_i - B, a_i]$; a B-only bus has $q \in [b_j - B, 0]$. These endpoint conditions keep inventory in $[0, B]$ before and after the remaining service. A monotone action between those inventories is also feasible throughout the early window. Terminal demand is that bus's service consumption minus q, between zero and B.

All upper endpoints are nonnegative and sum to P. The sum of positive lower endpoints is R, while the sum of all lower endpoints is $T = S - N_m B$. Thus total possible negative excursion is $R - T$. For $E \ge R$, set negative coordinates to zero and distribute purchases among the independent nonnegative intervals, whose sum spans $[R, P]$. Gross energy is then E. For $T \le E < R$, keep mandatory positive coordinates at their lower endpoints and distribute discharge of magnitude $R - E$ among the remaining negative intervals. This is possible exactly because $R - E \le R - T$ and has gross energy $2R - E$.

Conversely, total positive purchases are at least R, so any allocation with net E has gross energy at least $2R - E$ and at least E. The constructions attain both regimes in (8). Additional charge/discharge cycles cannot reduce gross energy and are unnecessary. □

Proof of Theorem 8. The lemma and early gross budget give $T \le E \le P$, $E \le K$ and $E \ge 2R - K$. All buses are present for the full early window; their monotone actions serialize in at most one hour when $K > 0$. For $K = 0$, the inequalities force $R = 0$, $E = 0$ and zero gross energy, without division. Every bus's terminal refill is nonnegative and at most B. They serialize at H exactly when $S - E \le H$; at $H = 0$ no positive terminal demand can be met. With a nonempty positive B batch, $E \le P < S$ makes that zero-terminal-power case infeasible.

For a fixed matching, the exact interval is consequently

$$[\max\{S - H, S - N_m B, 2R(M) - K\}, U].$$

Only $R(M)$ varies at fixed cardinality. Its attained minimum r_m gives the widest interval; all such intervals have common upper endpoint U. Their union is (10). Since r_m is nondecreasing, its threshold minimum is t_0; feasibility is precisely $t_0 \le U$, attained by a standalone fleet when needed.

Standard allowed-edge bipartite matching convexity gives discretely convex r_m. Interpolate it linearly on $[0, \nu]$. Taking the maximum of the resulting convex function $2r - K$, the affine function $S - (2n - m)B$ and the constant $S - H$ gives a convex function whose integer values are t_m. Each component is nondecreasing, so the thresholds are also nondecreasing. Zero increments can occur only in an initial block. After retaining its cheapest cardinality, the positive threshold increments are nondecreasing. Frontier slopes $-f/(t_m - t_{m-1})$ are therefore nondecreasing, followed by zero. Each endpoint is physical and each edge is its whole-plan mixture; every physical branch lies above this convex graph. This proves the lower-boundary assertion without introducing nonexistent or resource-infeasible cardinalities. □

Proof of Corollary 9. Strict convexity of G gives one minimizing hull load. At a retained threshold or on the final horizontal segment, a physical plan attains the frontier. In an adjacent open threshold interval, $f > 0$ makes its interpolated cost strictly below every physical branch at that load. The compact finite union of physical intervals then has a strictly larger minimum. The classical support criterion applies only with the stated signed-domain and price qualifications. At fee zero the empty matching physically attains the entire interval at zero operating cost. □

5.2 Identical paid idle buses

An idle bus has the same capacity, initial full inventory, terminal-full requirement, connector access and fee as a service bus. Its early signed charge lies in $[-B, 0]$ and its terminal refill is $-q$. It has no separate energy source or export tariff.

Proposition 10 (Idle buses under a total fleet cap)**.** *Allow identical paid idle buses, with total fleet at most $2n$. The complete cost-labelled projection in Theorem 8 is unchanged.*

Proof. Consider a matching of size m with k idle buses. Its total fleet is $N = 2n - m + k$, so the cap implies $k \le m$. The box argument of Lemma 7 still applies: idle buses add only negative headroom. The matching's interval is therefore

$$[\max\{S - H, S - NB, 2R - K\}, U],$$

at operating cost fN. Delete any k matching edges and use the resulting $m - k$ matching without idle buses. This fleet has the same size N and fee. Deleting edges cannot increase R, while the inventory term $S - NB$ and upper endpoint U stay fixed. Its feasible interval contains the original interval. Thus every projected point with idle buses has an identical load and cost without them. The reverse inclusion is immediate. □

For the two examples below, $H = S$. Even if the fleet cap is removed, fleets larger than $2n$ cannot improve the lower frontier: terminal capacity forces $E \ge 0$, and the $2n$ standalone buses realize every $E \in [0, U]$ at fee $2nf$. Any larger fleet has the same upper endpoint U and a higher fee when $f > 0$. It is therefore dominated at each feasible load. The literal projection acquires higher-cost layers, but its lower frontier and support conclusions are unchanged. This argument does not apply when $H > S$: additional idle inventory can then enable negative net early loads and may be useful.

5.3 Persistence and removal of an intermediate charge requirement

For the earlier anchor with A energies $(5, 15)$, B energies $(14, 15)$, battery capacity 20, $K = 15$ and $H = 49$, the matching values are $(r_0, r_1, r_2) = (0, 0, 9)$ and the new thresholds remain

$$(t_0, t_1, t_2) = (0, 0, 9).$$

Although individual buses gain discharging options, the entire cost-labelled projection is unchanged. At the earlier fee and supply objective, its previous positive gap therefore survives analytically. The terminal bound here enforces $E \geq 0$, so this particular conclusion does not require evaluating the old supply function at negative fleet loads. This is a proof consequence, not a new V2G implementation qualification or experiment.

Bidirectional charging can also remove a positive gap. Take complete crossmatching, A energies $(1, 1, 19)$, B energies $(2, 10, 19)$, capacity 20, $K = 2$, $H = 52$ and fee $f = 1$. Here $S = 52$ and $(r_0, r_1, r_2, r_3) = (0, 0, 0, 1)$. In the charge-only model, the cheapest physical operating cost is 4 on $0 \leq E < 1$ and 3 on $1 \leq E \leq 2$. Its hull boundary is $4 - E$ on $[0, 1]$, followed by the constant 3. For the convex supply cost

$$F(E, L) = E^2 + \frac{E + L}{5}, \qquad G(E) = E^2 + \frac{52}{5},$$

the physical minima at $E = 0$ and $E = 1$ tie, whereas the hull minimum is at $E = 1/2$. Consequently,

$$D = \frac{72}{5}, \qquad \mathrm{CH} = \frac{283}{20}, \qquad D - \mathrm{CH} = \frac{1}{4}.$$

With bidirectional charging, all four thresholds t_m are zero, so the three-bus operating cost 3 is physical throughout $[0, 2]$. In particular, pair the services as $(1, 19)$, $(1, 10)$ and $(19, 2)$ and assign signed early charges $0, -1, 1$, respectively. Their net load is zero and gross throughput is 2. Their terminal refills are $20, 12, 20$, totaling 52; all intermediate inventories respect capacity. Thus

$$D = \mathrm{CH} = 3 + \frac{52}{5} = \frac{67}{5}, \qquad E^* = 0.$$

At the own-load price $\nabla F(0, 52) = (1/5, 1/5)$, every feasible plan pays $52/5$ for net energy, and the three-bus plan minimizes operating cost. Hence it is supported under the stated settlement rule. Discharge from another assigned bus offsets a mandatory positive early purchase, using the gross connector budget in both directions. These exact values follow analytically from the projection theorem; they are not a numerical implementation qualification.

The result concerns lossless common-window physics and the specified net settlement. It neither establishes novelty of V2G or matching theory nor covers degradation, asymmetric prices, realistic travel, or the public 37-service case. That case remains charge-only and unchanged. No numerical qualification of this bidirectional extension is claimed.

6 Two services: capacity, fees and replenishment

The case $n = 1$, $a_1 = b_1 = 15$ gives a direct picture of the preceding threshold geometry. Varying its terminal resource also permits a complete connector-capacity characterization. Two mandatory depot-to-depot services consume 15 kWh each: A occupies $[0, 1]$ and B occupies $[2, 3]$. At most two identical buses are available, each with 20 kWh capacity, zero reserve and full initial inventory. Early charging is available only in $[1, 2]$ at 10 kW, both per bus and in total. Terminal charging is

available only in $[3, 4]$, using one connector of power $K \le 30$ kW. Every used bus must finish full at hour 4. Charging is continuous and lossless, with no taper or switching time. Travel cost, movement consumption and auxiliary load are zero. All prices and costs in this section are synthetic.

There are exactly two service partitions up to vehicle labels. With one bus, inventory after A is 5; it needs at least 10 early kWh to complete B. The early limit therefore fixes load $(10, 20)$. With two buses, A's bus buys $x \in [0, 10]$ early; terminal purchases are $15 - x$ for A's bus and 15 for B's bus. Each bus returns full. Both partitions buy exactly 30 kWh; initial inventory cannot subsidize the horizon.

Let $f \ge 0$ be the cost of a used bus and define

$$F(x, z) = 4x + \frac{x^2 + z^2}{10}, \qquad x, z \ge 0, \qquad g(x) = F(x, 30 - x) = 90 - 2x + \frac{x^2}{5}. \tag{11}$$

The early/late distinction permits different marginal supply costs despite constant total energy.

Theorem 11 (Nominal failure)**.** *For $K = 30$ and $f = 7$,*

$$D = 97, \qquad \mathrm{CH} = \frac{7591}{80} = 94.8875, \qquad \Delta = \frac{169}{80} > 0.$$

The unique physical planner load is $(10, 20)$ with one bus. Its own-price regret at $p = (6, 4)$ is 13.

The one-bus objective is $7 + g(10) = 97$; the two-bus branch minimizes $14 + g(x)$ at $x = 5$, giving 99. The hull instead mixes one bus at early energy 10 with two buses at early energy 0. Its cheapest operating cost at mean early energy x is $14 - 0.7x$. Thus its objective is

$$14 - 0.7x + g(x) = \frac{7591}{80} + \frac{1}{5}(x - 27/4)^2.$$

The mixture weights are $27/40$ and $13/40$. Both components are physical; the mean operating cost $371/40$ is a convexification. Running those plans on different days would have average physical cost 99.275, not 94.8875.

At the physical optimum's marginal price $(6, 4)$, its bill plus operating cost is 147. The two-bus zero-early alternative costs 134 at that fixed tariff. This 13-unit incentive changes the entire fleet. By contrast, the hull price $(107/20, 93/20)$ makes the two hull components tie, but is not the marginal supply price of the one-bus physical load. The distinction explains why a hull price and an own-load tariff answer different questions.

6.1 An exact connector and fee characterization

Using the same clamp notation, the next result describes the entire declared parameter domain, including its boundaries.

Theorem 12 (Connector capacity and vehicle fee)**.** *For $0 \le K \le 30$ and $f \ge 0$, the model is infeasible if $K < 20$. If $K = 20$, $D = \mathrm{CH} = f + 90$; both service partitions tie only at $f = 0$. For $K > 20$, put*

$$\ell = 30 - K, \quad d = K - 20, \quad t = \max\{5, \ell\}, \quad h = \mathrm{clamp}\left(5 + \frac{5f}{2d}; \ell, 10\right).$$

Then

$$D = \min\{f + 90,\ 2f + g(t)\}, \tag{12}$$

$$\mathrm{CH} = 2f - \frac{f(h - \ell)}{d} + g(h). \tag{13}$$

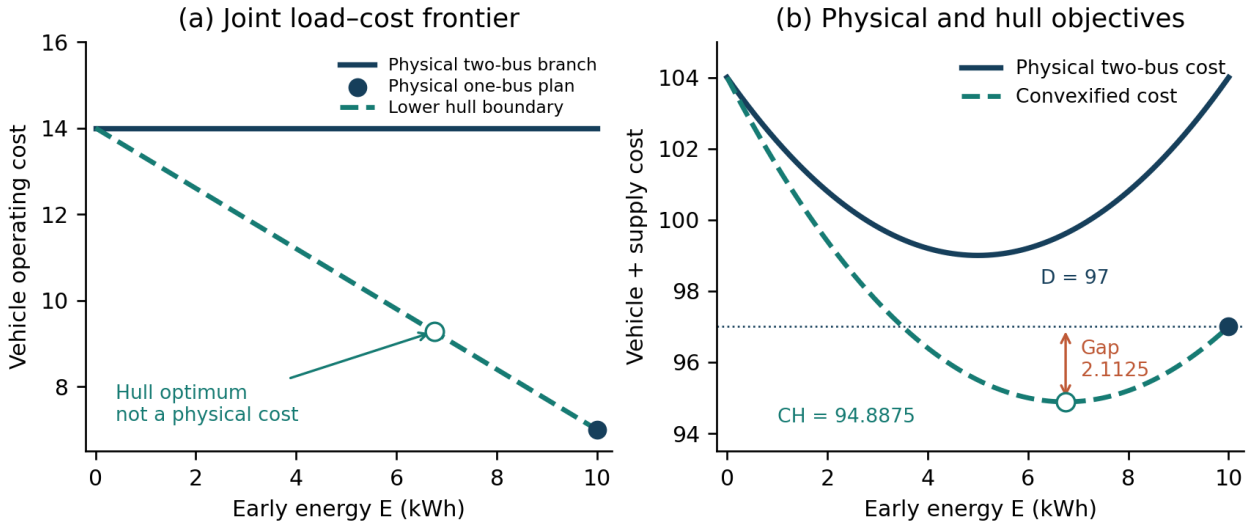


Figure 1: The nominal model's obstruction lies in the joint load–vehicle-cost frontier. Every early load from 0 to 10 is feasible with two buses, but the lower interpolated operating cost is not a physical fleet cost at an interior load. Adding the convex supply cost makes that distinction consequential. Curves illustrate the proved formulas; they are not sampled experimental data.

For $f > 0$, the gap is strictly positive exactly when

$$\max\left\{0, \frac{2}{5}d(\ell - 5)\right\} < f < 2d. \tag{14}$$

For $f = 0$ the gap is zero. Both physical planner partitions are retained when $f = 90 - g(t)$.

The proof is a physical projection and one-dimensional convex minimization, not a finite parameter scan (Appendix C.2). In particular, with $f = 7$ the gap is zero for $20 \le K \le 47/2$ and positive for $47/2 < K \le 30$. Extra charging capacity can create a price-support failure by enlarging the alternative set; it does not increase the planner's minimum cost or show that installing a connector worsens welfare.

Figure 2 contrasts this capacity effect with the equipment extension proved in Section 7. The support boundaries are included as weak-support cases; shaded positive-gap regions are strict interiors.

The full price-response value is

$$V(p) = \min\{f + 10p_1 + 20p_2,\ 2f + \ell p_1 + (30 - \ell)p_2,\ 2f + 10p_1 + 20p_2\}. \tag{15}$$

An affine objective attains its minimum on each physical segment at an endpoint. If $p_1 = p_2$, its whole two-bus interval is minimizing within that partition and must be retained whenever its bill also ties globally. This includes the zero-fee boundaries; an endpoint representative alone would omit part of the response set.

6.2 Positive reserve and charging losses

The nominal early limit is binding: adding reserve or losses without changing power can remove the one-bus branch. This is a feasibility issue, not disappearance of the pricing mechanism. A nearby

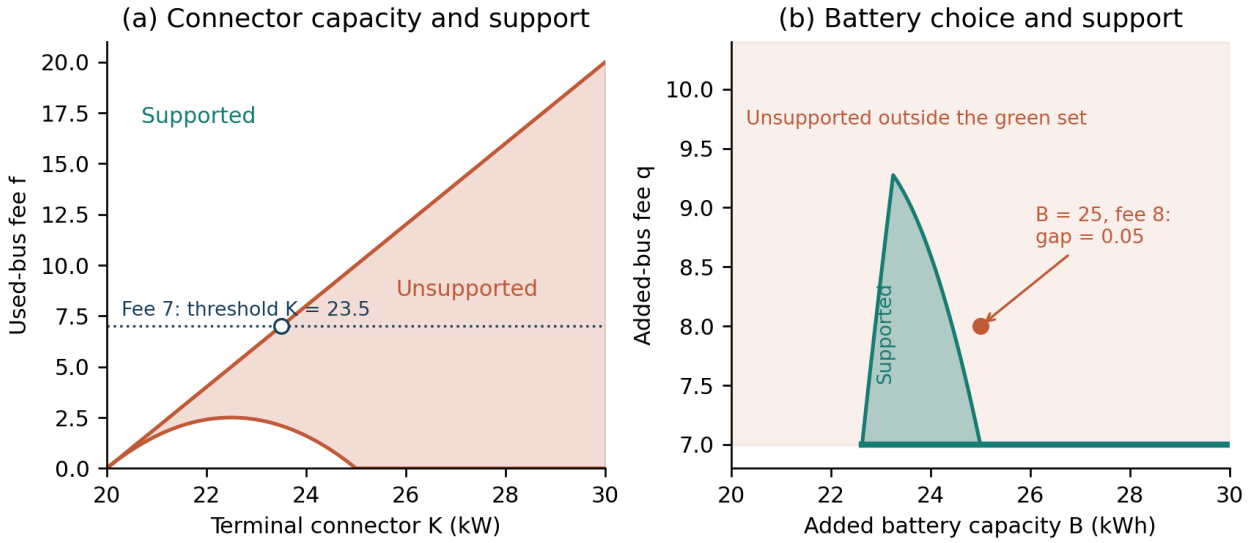


Figure 2: Analytical support regions. Left: the nominal model is feasible for $K \geq 20$; the orange interior has a positive physical–hull gap. At fee 7, support is lost only for $K > 23.5$. Right: green denotes support after adding battery capacity B at fee q while retaining the original buses. The green boundary is included. Better equipment alone is insufficient: $B = 25, q = 8$ still gives gap 0.05. Units and coefficients are synthetic.

construction uses reserve 1 kWh, efficiency 19/20, early power 12 kW and one terminal connector of 30 kW, keeping the service energies, capacity 20 and fee 7. The one-bus state sequence is

$$20 \longrightarrow 5 \longrightarrow 16 \longrightarrow 1 \longrightarrow 20.$$

It buys 220/19 grid kWh early and 20 terminally. A physical two-bus endpoint buys 30/19 early and 30 terminally; terminal sessions of 9/19 and 10/19 hour fit serially.

Proposition 13 (A nondegenerate neighborhood)**.** *For these modified parameters,*

$$D = \frac{38527}{361}, \qquad \mathrm{CH} = \frac{2987911}{28880}, \qquad \Delta = \frac{94249}{28880} > 0.$$

With terminal power 30 fixed, a positive gap persists in an open neighborhood of the stated reserve, efficiency and early power.

Appendix C.3 states the strict inequalities defining this neighborhood. The result remains conditional on constant efficiency, zero switching time and the absence of charging taper.

7 Does failure survive additional alternatives?

Let X be an old set of complete plans, Y a set of additions, and $Z = X \cup Y$. Use the same supply function and compact projected-image assumptions as before, with attained hull-dual prices where needed. Write D_A, CH_A, V_A for the values associated with a set A. Suppose $\Delta_X > 0$. The following distinction is essential: a retained old plan cannot be rescued by more deviations, but a newly admitted plan can become a supported optimum.

Proposition 14 (Expansion and an old dual certificate)**.** *For every old x and unchanged own-load price p,*

$$r_Z(x;p) \geq r_X(x;p) > 0.$$

Erasure of global failure therefore requires a new supported plan and, necessarily, $D_Y \leq \mathrm{CH}_X$. A sufficient condition for survival is $\min\{D_X, D_Y\} > \mathrm{CH}_X$.

Fix an attained old hull-dual optimizer p^*. If every added plan satisfies

$$c(y) + (p^*)^\top e(y) \geq V_X(p^*), \tag{16}$$

then $\mathrm{CH}_Z = \mathrm{CH}_X$. Under this condition, failure is erased exactly when some added plan attains equality in (16) and also satisfies $p^* \in \partial F(e(y))$. If (16) holds with a uniform margin $\varepsilon > 0$, then

$$\Delta_Z \geq \min\{\Delta_X, \varepsilon\} > 0.$$

These are elementary consequences of the classical dual certificate (Appendix D.1). They provide a useful way to ask whether a restricted physical mechanism survives genuinely richer alternatives. With strictly convex supply, supply equality at p^* fixes the old hull-optimal load; fleet equality then fixes its operating cost. Under (16), erasure means physically implementing that projected hull optimum. Without (16), the hull can move, and comparison with its old value is not enough.

7.1 Movement additions under unchanged resources

In the nominal model of Section 6, retain all original plans and allow additional movements consuming nonnegative energy and incurring nonnegative extra operating cost. Preserve the two services, at most two buses with 20 kWh batteries, fee 7, full initial and terminal inventories, the early 10 and terminal 30 kW power limits, charging windows and access. There is no energy recovery, inventory transfer or new charging opportunity.

Theorem 15 (Movement-only stability)**.** *Every feasible added plan is componentwise dominated in purchased-energy and operating-cost space by an old physical plan. Consequently*

$$D_Z = 97, \qquad \mathrm{CH}_Z = \frac{7591}{80}, \qquad \Delta_Z = \frac{169}{80}.$$

To see the physical reason, any added plan buys $30+w$ total energy, $w \geq 0$, and at most 10 early kWh. A two-bus plan with early purchase u is dominated by the old two-bus plan $(u, 30-u)$ of cost 14. A one-bus plan must still obtain at least 10 early kWh to complete both services; the early limit forces $u = 10$, and the old one-bus plan $(10, 20)$ of cost 7 dominates it. The same replacement works inside every hull mixture because F is coordinatewise nondecreasing. The theorem therefore preserves both values, not merely positivity of their difference.

This is an unchanged-resource result. The original no-movement construction already includes both possible service partitions and every continuous charging choice. A new projected point cannot be called an overlooked feasible duty without explaining which physical assumption admits it.

7.2 An explicitly enlarged battery choice

Now change the equipment. Keep buses with 20 kWh batteries and fee 7, and add buses of capacity $B \in [20, 30]$ and fee $q \geq 7$. Two of each type are available, but at most two buses total are used. Every used bus starts and ends at its own full capacity. All service and charging constraints are unchanged.

Put $a = 30 - B$. The new one-bus branch buys $x \in [a, 10]$ early and $30 - x$ terminally, with operating cost q. Every two-bus type assignment permits the old interval $[0, 10]$ and costs at least 14; the old–old assignment therefore dominates mixed and new–new assignments in projected economic comparisons. Their physical records and equal-fee ties remain part of the complete alternative set.

Theorem 16 (Battery and fee conditions for erasure)**.** *For $20 \le B \le 30$ and $q \ge 7$, let $a = 30 - B$ and*

$$A(a) = 14 + 2a - \frac{2}{5}a^2, \qquad C(a) = 7 + (10 - a)\left(\frac{2}{5}a - 2\right).$$

The enlarged model has a supported physical plan exactly when

$$(a \le 5 \text{ and } q = 7) \quad \text{or} \quad (a > 5 \text{ and } 7 \le q \le \min\{A(a), C(a)\}). \tag{17}$$

The inequalities include weak support at ties. At fee $q = 7$, this becomes

$$B \ge \frac{55 - \sqrt{95}}{2} \simeq 22.6266. \tag{18}$$

Outside (17), the physical–hull gap is positive.

The new branch's physical minimizer is $t = \max\{5, a\}$. If $a \le 5$, its marginal prices at $t = 5$ are equal; the new plan minimizes its bill against the cheaper old one-bus plan only at fee 7. If $a > 5$, the two inequalities in (17) compare it respectively with the old zero-early two-bus plan and old one-bus plan. The proof in Appendix D.2 checks the entire continuous opportunity set.

The case $B = 25$ shows why a new plan's physical cost is not the whole story. For $q \ge 7$,

$$D_Z = \min\{97, 85 + q\}, \tag{19}$$

$$\mathrm{CH}_Z = \begin{cases} 85 + q - (q - 7)^2/20, & 7 \le q \le 21/2, \\ 7591/80, & q \ge 21/2. \end{cases} \tag{20}$$

At fee 7, the new one-bus load $(5, 25)$ has total cost 92 and is supported at $(5, 5)$. At fee 8, its physical cost 93 is below the *old* hull value 94.8875, but the *new* hull value is 92.95. Its regret is 1 and the gap remains $1/20$. Thus the necessary condition $D_Y \le \mathrm{CH}_X$ in Proposition 14 is not sufficient.

At fee 12, the old one-bus load $(10, 20)$ and new one-bus load $(5, 25)$ both have physical cost 97. Their own-price regrets are 13 and 5, respectively; $\mathrm{CH}_Z = 7591/80$. Reporting a single selected planner optimum would hide this incentive difference. The new equipment can erase failure, but its presence alone does not do so.

8 Aggregation and the owner of the deviation

The homogeneous specialization of Section 4 provides a separate way to distinguish a planning-value gap from a participant's incentive. Replicate the nominal services to n simultaneous copies of A and n copies of B. Retain the per-bus physical parameters and fee 7. Provide n early connectors of power 10 kW and n terminal connectors of power 30 kW. Supply scales with the market:

$$F_n(E, L) = 4E + \frac{E^2 + L^2}{10n}. \tag{21}$$

This scaling keeps the marginal-price range bounded; it is an assumption of the family, not an asymptotic claim for an unchanged supply curve.

If m buses each cover an A–B pair, the remaining services use $n-m$ A-only and $n-m$ B-only buses. Operating cost is $14n-7m$, early energy is in $[10m, 10n]$, and terminal energy is $30n-E$. All terminal charges fit: each paired bus takes one connector, while each unpaired A/B pair shares one connector serially. Thus the projected opportunities used below have explicit physical realizations. All A-to-B crossmatchings are admitted in this aggregate model. Identical energies make their projected outcomes depend only on m; fixed pair duties are not being imposed here. In the threshold notation, $r_m = 10m$.

Theorem 17 (Planning gap and whole-owner regret)**.** *The hull value is* $\mathrm{CH}_n = 7591n/80$*. Every physical planner optimum takes an integer* m *nearest to* $27n/40$*, retaining both choices at a tie. Writing* $\delta = m - 27n/40$*,*

$$\Delta_n = \frac{20\delta^2}{n} \le \frac{5}{n}. \tag{22}$$

For one owner controlling all services and buses, the own-price regret is

$$r_n = \begin{cases} m\,40\delta/n, & \delta \ge 0, \\ (n-m)(-40\delta/n), & \delta < 0. \end{cases} \tag{23}$$

In particular, along $n = 40k+1$*,*

$$\Delta_n = \frac{169}{80n} \longrightarrow 0, \qquad r_n = \frac{351}{40} + \frac{169}{40n} \longrightarrow \frac{351}{40}.$$

For multiples of 40, both quantities are zero.

Approximate convexity of an aggregate does not by itself bound a composite owner's absolute incentive to replace its entire aggregate. This is consistent with aggregation perspectives [17, 7]. In this family regret per bus and regret relative to physical cost vanish; the nonvanishing statement concerns the whole owner's *absolute* fixed-price gain along a declared subsequence.

Now divide ownership among n operators, each responsible for one A/B pair, with a reserved early connector and a reserved terminal connector. Each may use one or two buses and choose its two-bus early charge in $[0, 10]$. Those reserved rights ensure unilateral deviations remain feasible with every other participant's plan held fixed. An aggregate optimum may pair A and B copies across these ownership labels. Before making an individual comparison, implement its same projected outcome by assigning m owners a one-bus plan and the remaining $n-m$ owners a two-bus plan on their own reserved service pairs. Identical service copies and the explicit connector assignment make this possible with unchanged aggregate load, operating cost and marginal price. This is a reimplementation of the economic outcome, not a claim that a cross-matched literal schedule already respects the newly imposed ownership rights. This step is specific to identical services. In the heterogeneous example of Section 4, fixed pairs with energies $(5, 14)$ and $(15, 15)$ require 10 early kWh when both use one bus, whereas the crossed complete assignment requires only 9. The latter outcome cannot be transferred to that fixed-pair institution by relabeling. No individual regret bound for heterogeneous fixed ownership is asserted below.

Corollary 18 (Reserved participants)**.** *For any physical planner optimum reimplemented in the reserved-pair institution as above, each reserved pair's regret at its marginal price is at most* $20/n$*. The regrets sum to* (23)*. Along* $n = 40k+1$*, each dissatisfied one-bus owner can gain only* $13/n$*.*

The proof is the same per-pair bill comparison (Appendix E). At $n = 20$ the two tied physical planner optima with $m = 13$ and $m = 14$ have whole-owner regrets 7 and 14, respectively. Neither the whole-owner calculation nor the reserved-pair calculation specifies an institution for unreserved shared chargers or strategic anticipation of a tariff reset.

9 What the public timetable does and does not establish

The analytical construction proves possibility, not operational frequency. The public case uses the Hildenbrand timetable from the Sistig data [15, 16]: 37 mandatory services and the complete directed deadhead matrix. Depot 15 and depot 16 define separate one-depot modifications of the original network. Source timetable offsets, including next-day services, are retained; source data do not provide the synthetic energy and cost assumptions below.

Service consumption is 1.20 kWh/km plus 16 kWh per service hour; directed deadheads use 0.96 kWh/km plus 16 kWh per travel hour, and explicit off-depot waiting uses 16 kWh/h. Depot-parked auxiliary draw is zero. A used bus has 400 kWh usable inventory, zero reserve, unit efficiency and 360 kW acceptance. One 360 kW connector is shared. Every used bus starts full and returns full by a synthetic 30:00 service-block boundary. There is no taper or switching time. Monetary movement cost is zero; movement time and energy remain physical constraints. The used-bus fee is 100. This finite block is not a certificate of repeating daily operation.

The initial flat tariff is 0.2 per kWh. Because the complete-plan objective is linear, its physical and hull optima have equal value regardless of the discrete assignments. A native numerical flat optimum near 408.53 at depot 15 is tolerance-qualified and does not identify the ideal stored-input rational optimum. Exact rational evidence instead provides a two-bus physical witness and a cardinality-gated path-cover relaxation, giving outward bounds

$$D_{15}^{\mathrm{flat}} \in [404.92, 408.54].$$

An exact one-bus obstruction establishes minimum fleet size two. Depot 16 has an exact lower bound 414.39, but its reported two-bus upper witness is numerical; the resulting interval must be labeled mixed evidence.

On the same finite depot 15 ideal model, let

$$F(e) = \sum_{t=1}^{30} \left(0.2 e_t + \frac{e_t^2}{1800} \right). \tag{24}$$

The displayed coefficients denote the original stored binary values in the exact certificates.[1] A new exact two-bus witness and an independently reconstructed rational lower certificate for a per-leg perspective energy-flow relaxation give the presently established outward bounds

$$434.17 \le \mathrm{CH}_{15} \le D_{15} \le 441.54, \qquad 0 \le D_{15} - \mathrm{CH}_{15} \le 7.37. \tag{25}$$

The new exact lower is approximately 434.17948, rounded downward here; its validity follows from a supply tangent, sign-valid rational row multipliers and finite-bound compensation for stationarity residuals. It improves the earlier energy-floor lower of 429.22. The continuous energy-flow relaxation is not asserted to equal the complete-fleet hull. A separate native integer solve returned OPTIMAL with objective 441.532912…, but its candidate failed strict physical replay and did not itself improve the exact upper or establish ideal-model optimality. The native status and the independently checked rational lower are different evidence. An earlier admitted witness had cost 442.123049…. A time-limited classical MIQP supplied its routes and charging proposal; two terminal energy changes smaller than $7 \cdot 10^{-12}$ kWh restored exact full inventories. A separate rational event replay then verified all services, directed movements, battery bounds, shared connector use

[1] Precisely, the linear coefficient is 3602879701896397/18014398509481984 and the quadratic Hessian coefficient is 5124095576030431/4611686018427387904; the quadratic term is half the latter times e_t^2. The familiar decimal expression displays their intended scale.

and full replenishment. The original unadmitted numerical proposal is retained separately. No route or nonterminal charging allocation changed.

For any fleet with at least three buses, its operating cost and only the linear mandatory-service-energy component already exceed 477.82. Movement consumption and the quadratic cost are nonnegative. Such a fleet cannot improve the new physical upper. Together with the one-bus obstruction, this establishes exactly two buses at any physical planner optimum. It does not exclude larger-fleet components from arbitrary hull mixtures or from a price-response problem at another tariff.

Separately, a second strictly admitted physical plan and the protected witness give a feasible complete-plan mixture with value $441.600681\ldots$, hence $\mathrm{CH}_{15} \leq 441.61$. Its negative initial directional derivative proves that the protected witness is unsupported at its own marginal prices. A mixture is not a dispatch, and an unsupported incumbent is not proof that an optimal physical fleet is unsupported. A subsequent fixed-route charging QP returned the same lower numerical value $441.532912\ldots$, but its first exact dual recovery failed on a multiplier sign; the conditional global own-price checks were therefore unattempted. A distinct rational phase-I method then constructed a new exact primal, admitted it by independent physical replay, and found sign-valid multipliers certifying its fixed-route charging optimum. This supplies the stronger physical upper in (25); it also supersedes the earlier mixture upper as a bound on CH. The original failure remains recorded. Fixed-route optimality still does not exclude a profitable complete-fleet deviation or certify global marginal-price support. A subsequent full declared-graph response query at this stronger incumbent's exact marginal prices returned a native integer optimum but again failed strict physical admission. Its value alone is not a profitable physical deviation. Using only those selected route identifiers, one new linear charging problem and rational construction produced a separately admitted response. Its fixed-price objective has an exact fixed-route optimality certificate. Its nonlinear planning cost is higher, but its bill at the incumbent's posted prices is strictly lower. Thus this stronger incumbent is also provably unsupported. A directly checked rational mixture of the two admitted plans has a value strictly below the incumbent's physical cost. The mixture is a feasible hull upper, and its exact value is recorded in the reproducibility artifacts; it is not a physical schedule or a global hull optimum. No full-route discovery or old physical replay was repeated to construct it.

The bounds neither determine D_{15} or CH_{15} nor establish a positive gap. No new public sign certificate is asserted in this draft.

There are distinct valid ways to close this question. An exact physical witness x and a globally valid linear-response lower bound matching $c(x)+\nabla F(e(x))^{\top}e(x)$ prove support and hence $D = \mathrm{CH}$. Alternatively, a physical lower bound strictly above a feasible mixture's hull upper bound proves positive gap. A globally certified hull optimizer can also prove failure through nonimplementability: strict convexity in (24) fixes its load and operating cost, and a hull operating cost not in $100\mathbb{Z}_{+}$ cannot be the cost of a physical fleet. All these routes require a global certificate or a complete infeasibility argument. A finite plan bank, incumbent regret, or a small native solver gap does not settle the sign of the ideal-model gap.

10 Implications and limits

The examples identify a conditional failure of a specified posted-tariff institution. The supply function is convex; the indivisibility lies on the fleet side. Continuous energy decisions and terminal energy accounting do not eliminate that indivisibility. Conversely, an indivisible fleet need not lack supporting marginal prices: the exact parameter regions include zero-gap cases, and an added

battery choice can physically implement a supported load. The correct claim is a possibility with characterized conditions, not inevitable failure of marginal tariffs.

The expansion results are deliberately explicit about resource changes. The movement-only stability theorem preserves charging access, equipment, service requirements and nonnegative movement consumption. It says nothing about a new charging location, regenerative energy, altered travel energy, service cancellation or a more capable battery. Those changes can alter the relevant energy projection. The equipment result studies one such change directly rather than describing an arbitrary projected point as an omitted feasible duty of the original model.

The participant definition is equally consequential. A whole-fleet deviation can replace every service assignment. Independent operators need compatible resource rights before their unilateral alternatives are well defined. The replication theorem uses reserved service pairs and connectors to make that comparison exact. It does not infer incentives for independently owned buses competing for the same unreserved depot connector. Heterogeneous fixed service pairs cannot generally reproduce the efficient complete-crossmatching outcome by relabeling; that step in the replication comparison relies on identical services.

All analytical models omit stochastic delays, battery degradation, temperature effects, charging taper and switching time. Monetary and supply coefficients are synthetic. The positive-gap neighborhood admits a reserve, constant losses and one connector, but remains conditional on those abstractions. The analytical formulas and exact implementation comparisons support the declared mechanisms, not estimates of an operator's savings or an assessment of a deployed market design.

The two-batch theorem allows fixed shared charging resources but assumes two simultaneous service batches and no travel. The public timetable has varied service times, movement and waiting energy, and one shared connector, so it is outside that theorem's scope. Extending the projection to such coupled resources or obtaining a complete public certificate would connect the structural analysis to a wider operational setting. The present bounds leave that connection unresolved.

11 Conclusion

An electric-bus fleet can lack marginal-price support even when charging is continuous and every used battery is fully replenished. For heterogeneous two-batch services with allowed successor matchings and positive vehicle fees, cardinality-matching thresholds determine exactly which hull-optimal loads are physically supported. Bidirectional charging preserves that structure with different thresholds: it can leave a positive gap intact or eliminate it. Gross connector time and terminal replacement remain binding even when net exports are allowed. The two-service model shows how connector capacity, vehicle fees and battery options change that answer. Energy-consuming movement additions preserve the obstruction under unchanged resources; a sufficiently capable battery can remove it at an appropriate fee. Neither a cheaper newly admitted plan nor a vanishing planning gap alone resolves the relevant incentive question. The public case remains a certification problem rather than empirical confirmation of the mechanism.

Evidence and reproducibility

The analytical claims concern the stated ideal models. Exact rational implementation comparisons use independently formulated physical enumeration and analytical formulas; agreement is an internal cross-check, not external replication. The connector/fee qualification contains twenty fixed cases, and the support-extension qualification fourteen fixed cases, with explicit boundary and corruption controls. These cases qualify implementations; the proofs establish the continuous-domain

statements. The heterogeneous adequate-resource specialization matched independent raw SOC enumeration and threshold formulas on all eight fixed cases; eighteen negative controls and a smoke anchor passed, including tie and continuous-response checks. The fixed single-shared-connector extension separately matched raw SOC enumeration and serial witnesses against the sorted-threshold formulas on all fourteen prospectively fixed cases, with twenty-four negative controls and a smoke anchor passing. These include physical collinear-threshold and clipped lower-boundary optima, assignment ties and response continua. Both are internal qualifications, not external replication; the eight-case run concerns the separate n-connector resource policy. Archived outcome and resource records distinguish exact arithmetic from tolerance-qualified native computations. This evidence update reads stored qualification results without rerunning the scientific calculations. No public optimal-fleet support conclusion is drawn from the finite plan bank or solver statuses.

A A separate adequate-resource specialization

We next study a different hardware policy in which the complete assignment set has an exact energy projection. There are $n \geq 1$ mandatory depot-to-depot services A_i occupying $[0, 1]$ and n mandatory services B_j occupying $[2, 3]$. Their consumptions are a_i and b_j, with

$$0 < a_i \leq 20, \qquad 0 < b_j \leq 20, \qquad a_i + b_j \leq 30 \quad \text{for every } i, j. \tag{26}$$

Every A-to-B assignment is allowed, without movement time, energy or cost. At most $2n$ identical buses may be used. Each has capacity 20 kWh, starts full and returns full by hour 4. Its operating fee is strictly positive, $f > 0$.

Early charging in $[1, 2]$ uses n connectors of 10 kW each and an aggregate grid limit of $10n$ kW. Terminal charging in $[3, 4]$ uses n connectors of 30 kW each and an aggregate grid limit of $30n$ kW. Per-bus acceptance matches those connector powers. There is no other charging, reserve, loss, taper or switching time. The grid limits are part of the assumption: connector counts alone would not establish the simultaneous allocation used below.

Since the services in each batch are simultaneous and have positive duration, a bus can cover at most one A and one B. Thus a complete assignment is a partial bipartite matching M. A matched pair uses one bus; unmatched services each use a separate bus. All crossmatchings and all cardinalities $m = 0, \ldots, n$ are retained.

Define

$$S = \sum_i a_i + \sum_j b_j, \qquad u_i = \min\{a_i, 10\}, \qquad U = \sum_i u_i, \tag{27}$$

$$r_{ij} = [a_i + b_j - 20]_+, \qquad r_m = \min_{|M|=m} \sum_{(i,j)\in M} r_{ij}, \qquad c_m = f(2n - m), \tag{28}$$

where $r_0 = 0$. The edge cost r_{ij} is the minimum early purchase needed by a bus covering A_i then B_j. The thresholds r_m are minimum-cost cardinality-matching values, not samples from a plan bank.

Theorem 19 (Exact heterogeneous fleet projection)**.** *Under* (26) *and the stated resources, including $f > 0$, the complete physical projection is exactly*

$$\bigcup_{m=0}^{n} \{(E, S - E, c_m) : r_m \leq E \leq U\}. \tag{29}$$

The thresholds are nondecreasing and discretely convex:

$$0 \le r_1 - r_0 \le r_2 - r_1 \le \cdots \le r_n - r_{n-1}. \tag{30}$$

Let $m_0 = \max\{m : r_m = 0\}$. *The lowest operating-cost boundary of the complete-plan hull is the affine interpolant through* (r_m, c_m) *for* $m = m_0, \ldots, n$, *followed by the horizontal segment* c_n *on* $[r_n, U]$.

The physical part of the theorem is the battery and connector argument. A bus that has served A_i can buy any $x_i \le u_i$ early; pairing it with B_j adds the lower bound $x_i \ge r_{ij}$. There are exactly n A buses, so their individual charge intervals can be allocated simultaneously. Terminal replenishment fits by grouping each matched bus alone and each unmatched A/B pair together on one connector. The discrete-convexity step uses standard bipartite min-cost-flow integrality and convex LP value functions. It is not a new matching theorem. The complete proof is in Appendix B.

Use the same scaled supply family as the later replication calculation:

$$F_n(E, L) = 4E + \frac{E^2 + L^2}{10n}, \qquad G(E) = F_n(E, S - E), \qquad E_0 = \frac{S - 20n}{2}. \tag{31}$$

Strict convexity makes the hull minimizing early load E^* unique.

Theorem 20 (A complete support criterion for the two-batch class)**.** *For strictly positive fee* $f > 0$ *under Theorem 19,*

$$D = \mathrm{CH} \quad \Longleftrightarrow \quad E^* \in \{r_m : m = m_0, \ldots, n\} \cup [r_n, U]. \tag{32}$$

The gap is strictly positive exactly when E^* *lies strictly inside an adjacent threshold interval* (r_{j-1}, r_j), $j > m_0$. *If* $d_j = r_j - r_{j-1} > 0$, *an intermediate threshold* r_m *minimizes the hull objective exactly when*

$$\frac{f}{d_{m+1}} \le G'(r_m) \le \frac{f}{d_m}, \qquad m_0 < m < n. \tag{33}$$

The criterion includes weak support at ties. A threshold is physical, but an interior point between consecutive thresholds interpolates operating costs that no fleet at that early load can attain. Collinear adjacent segments do not erase the intervening physical threshold. If $f = 0$, the conclusion about strict interiors would fail: the zero-matching branch realizes every early load at zero operating cost, so the gap is always zero. Strict positivity of the fee is therefore substantive.

The values themselves require only matching thresholds and scalar convex minimizations. With $\mathrm{clamp}(v; \ell, u) = \min\{u, \max\{\ell, v\}\}$,

$$D = \min_{0 \le m \le n} \{c_m + G(\mathrm{clamp}(E_0; r_m, U))\}. \tag{34}$$

For each sloping hull edge $j > m_0$, evaluate its objective at

$$h_j = \mathrm{clamp}\left(E_0 + \frac{5nf}{2d_j}; r_{j-1}, r_j\right). \tag{35}$$

Compare these values with the minimum on the final horizontal segment. Their minimum is CH. If $m_0 = n$, that horizontal segment alone remains and $D = \mathrm{CH}$.

The separate eight-case exact qualification of this hardware policy matched raw SOC enumeration against its threshold formulas in all eight fixed cases, including assignment ties and continuous response sets. All eighteen negative controls and the smoke anchor passed. This is an internal

implementation qualification of the stated n-connector policy, not the fixed single-connector model or an operational prevalence estimate.

For the main-text example in (7), the adequate-resource policy also gives $U = 15$ and the same cost-labelled projection. Its zero-early endpoint has terminal demands 19, 15 and 15, fitting two 30 kW connectors by serially grouping the last two demands. Its crossed full matching buys 9 early on the 15-energy A bus and replenishes both buses by 20 terminally. These are witnesses for this separate hardware policy. The main text supplies new serial realizations at early power 15 and terminal power 49; the parallel witness timings here are not reused as single-connector schedules. Individual allocation sets can differ even when the joint load–cost projections coincide.

B Heterogeneous complete-assignment proofs

Proof of Theorem 4. Complete assignments are precisely partial matchings in the declared graph Γ, with every unmatched service retaining a separate bus. For an allowed matching M, a bus serving A_i has inventory $B - a_i$ before early charging. Headroom gives $0 \le x_i \le a_i$; pairing B_j adds $x_i \ge [a_i + b_j - B]_+ \le a_i$. An unmatched B bus is full throughout the early window. Summing these independent intervals gives $[R(M), \sum_i a_i]$. All A buses are present for the entire early window. At $K > 0$ serial charging takes E/K hours, so exactly $E \le K$ is required. When $K = 0$, only zero early energy is possible.

Every bus is present at hour 3. Its terminal demand is $a_i + b_j - x_i$ if paired, $a_i - x_i$ if A-only, or b_j if B-only. These quantities are nonnegative, individually no greater than B, and total $S - E$. At $H > 0$ serial replenishment takes $(S - E)/H$ hours, giving precisely $S - E \le H$. At $H = 0$ no positive terminal demand can be met. Monotone charging preserves inventory bounds and restores each bus full. Thus each matching has exactly interval $[\max\{L, R(M)\}, U]$ when nonempty. Finite minima and a common upper endpoint give the stated cardinality union for $0 \le m \le \nu$. Every such cardinality exists by deleting edges from a maximum allowed matching. Since the empty matching and every singleton option are retained, feasibility is exactly $L \le U$, also when $\nu = 0$ or some vertices are isolated.

Deleting an edge from a minimum size-m matching gives $r_{m-1} \le r_m$. For discrete convexity, use the standard unit-capacity source–A–B–sink network containing only the allowed A–B edges, with their costs r_{ij}. Its fixed-flow LP is integral at integer flow values $m = 0, \dots, \nu$. The LP value is convex in required flow by duality, so $2r_m \le r_{m-1} + r_{m+1}$ for $1 \le m < \nu$. This reasoning uses bipartiteness, not completeness or the sorted-energy shortcut. All thresholds at or below L are dominated in cost by m_0. The first positive retained horizontal increment is $r_{m_0+1} - L$, no greater than $r_{m_0+1} - r_{m_0}$; later increments are the original nondecreasing matching increments. Hence frontier slopes $-f/\text{increment}$ are nondecreasing, followed by zero. Endpoints are physical and their interpolants are hull mixtures. Every physical branch lies above this convex graph, so every mixture does too. The graph is therefore exactly the lower hull boundary. Cardinalities above $m_{\max}$ have empty physical intervals and supply no endpoint. Cardinalities above ν supply no matching at all. If $m_0 = m_{\max}$, including the empty-graph case, only the physical horizontal segment remains. □

Proof of Corollary 5. The convex frontier plus strictly convex continuous G has one minimizing load. At L, at a retained threshold or on the final horizontal segment, a physical plan attains the frontier. Strictly between consecutive retained thresholds, its operating cost is strictly below that of every physical branch because $f > 0$. No physical plan attains the hull minimizer; compactness of the finite union of physical segments gives $D > \mathrm{CH}$. This retains physical thresholds even on a larger collinear hull face. The mathematical value equivalence becomes a price-support equivalence only with the framework's separate qualifications. □

Proof of Corollary 6. Under the corollary's complete-graph assumption, replacing a selected service by an unused smaller energy while keeping its partner cannot increase the coordinatewise nondecreasing edge cost. Thus some optimal size-m matching uses the m smallest energies in each batch. Let $h(t) = [t - B]_+$. For $a \le a'$ and $b \le b'$, monotonicity of the convex increment $h(t + a' - a) - h(t)$ gives

$$h(a + b) + h(a' + b') \ge h(a + b') + h(a' + b).$$

Swapping any two partners ordered in the same increasing direction therefore cannot raise cost. Repeating exchanges yields opposite order and the stated sum. Ties and zero edge costs require no separate assumption. □

Proof of Theorem 19. Every bus serves at most one A and one B, so service assignments are exactly partial bipartite matchings. For a fixed matching M, let x_i be the early purchase on the bus serving A_i. After that service it has inventory $20 - a_i$. Capacity and early acceptance imply $0 \le x_i \le u_i$. If matched to B_j, its inventory after B is $20 - a_i + x_i - b_j$, giving $x_i \ge r_{ij}$. The assumptions imply $r_{ij} \le a_i$ and $r_{ij} \le 10$, hence each interval is nonempty. An unmatched B bus starts full and has no earlier consumption, so it cannot accept early charge. The independent early intervals sum to exactly

$$\left[\sum_{(i,j) \in M} r_{ij}, U \right].$$

There are n A buses and n early connectors, with aggregate grid power $10n$, so every choice of these purchases is realizable.

For terminal charging, a matched bus needs $a_i + b_j - x_i \le 20$ kWh and fits alone on one connector. An unmatched A bus needs $a_i - x_i$; an unmatched B bus needs b_j. Group unmatched A and B buses arbitrarily; each group's sum is at most $a_i + b_j \le 30$ and fits serially on one 30 kW connector during the terminal hour. There are m matched groups and $n - m$ unmatched groups. Aggregate power $30n$ allows all groups simultaneously. Every bus reaches its own full inventory without intermediate overcharge or energy deficit. Full replenishment fixes total purchase S.

All size-m matchings have the same operating cost and upper early endpoint U, and there are finitely many. Their interval union is $[r_m, U]$, proving the exact projection without restricting the matching set.

Deleting one edge from a minimum size-m matching proves $r_{m-1} \le r_m$. To prove discrete convexity, represent the bipartite matching as unit-capacity flow from a source through A and B vertices to a sink, charging edge cost r_{ij}. At integer required flow the LP is integral and has value r_m. Its value as a function of required flow is convex by LP duality, a supremum of affine functions. Therefore $2r_m \le r_{m-1} + r_{m+1}$.

Nonnegative nondecreasing increments can vanish only in an initial block. After m_0, set $d_j = r_j - r_{j-1} > 0$. The interpolant slopes $-f/d_j$ are nondecreasing, and the final slope zero is no smaller. The graph is convex and no higher than any physical horizontal branch; the earlier zero-threshold branches have strictly larger cost because $f > 0$. Each interpolant edge is a convex combination of physical endpoints, and the final horizontal segment is physical. Thus its graph belongs to the hull. Conversely every physical point lies on or above this convex graph, and therefore so does every convex combination. This proves it is exactly the hull's lowest operating-cost boundary. □

Proof of Theorem 20 and the value formulas. The hull boundary is convex; G is strictly convex with derivative $G'(E) = 2(E - E_0)/(5n)$. Their sum has a unique minimizing early load. At a threshold or on the final horizontal segment, a physical plan attains the hull boundary, so $D = \mathrm{CH}$. In an adjacent open threshold interval, the hull operating cost is strictly below the cheapest physical

branch c_{j-1}, since $f > 0$. No physical plan attains the unique hull-optimal projected point, and every other load has larger hull objective. The finite union of compact physical segments attains its minimum, proving $D > \mathrm{CH}$. Proposition 2 supplies the support equivalence.

At an intermediate threshold r_m, the objective's one-sided slopes are $G'(r_m) - f/d_m$ and $G'(r_m) - f/d_{m+1}$. Their signs straddle zero exactly under (33), including weak inequalities. Strict convexity on each physical branch gives (34). On a sloping hull edge its derivative is $G'(E) - f/d_j$, whose zero is $E_0 + 5nf/(2d_j)$. Clamping to the edge gives (35); the remaining horizontal segment minimizes $c_n + G$ directly. This covers the $m_0 = n$ case without dividing by a zero threshold increment. □

Details of the heterogeneous example. The stated matrix gives $r_0 = r_1 = 0$ and $r_2 = 9$. Physical minima for $m = 0, 1$ occur at $E = 9/2$ and cost $5841/40$ and $5561/40$, respectively. For $m = 2$, the minimum is the endpoint $E = 9$, with value $2681/20$. The sloping hull edge objective is

$$\frac{2821}{20} - \frac{151}{90}E + \frac{E^2}{10},$$

whose minimum is $434201/3240$ at $151/18$. Subtracting gives $\Delta = 121/3240$. The endpoint weights stated in the text realize the same load and operating interpolation. At $p = (49/10, 4)$, the price difference is positive, so each branch minimizes its bill at its lowest early endpoint. Their bills plus operating costs are 224, 217 and $2181/10$. Hence the full-set regret is $11/10$, not merely a gain against an arbitrarily selected alternative. All calculations are exact algebra on the displayed ideal model. □

C Support, physical projection and parameter proofs

C.1 The classical support criterion

Proof of Proposition 2. For any hull point (e, c), linear minimization gives $c + p^\top e \geq V(p)$. If $p \in \partial F(e(x))$, then $F(e) \geq F(e(x)) + p^\top(e - e(x))$. Adding and minimizing over the hull gives

$$\mathrm{CH} \geq V(p) + F(e(x)) - p^\top e(x) = c(x) + F(e(x)) - r(x; p).$$

This proves (4). If $r(x; p) = 0$, equality of physical and hull values follows immediately.

Conversely let $D = \mathrm{CH}$. Compactness and lower semicontinuity give a physical minimizer. The relative-interior qualification gives strong Fenchel duality and an attained maximum $\mathrm{CH} = V(p^*) - F^*(p^*)$. Under the alternative finite-minorant condition, apply the same duality theorem to the finite minorant. It agrees with F on hull loads and its conjugate is at least F^*. The corresponding price gives a dual bound for F at least CH; weak duality forces equality, hence attainment for F as well.

For any physical minimizer x and any attained hull-dual maximizer p^*,

$$\underbrace{c(x) + (p^*)^\top e(x) - V(p^*)}_{\geq 0} + \underbrace{F(e(x)) - (p^*)^\top e(x) + F^*(p^*)}_{\geq 0} = D - \mathrm{CH} = 0.$$

Both terms vanish. Fenchel equality gives $p^* \in \partial F(e(x))$; the first term gives support. For restricted economic prices the converse requires an attained dual price in that subset, as stated. Every dual maximizer satisfies the displayed unrestricted algebraic conclusion; only the admissible maximizers give support under a restricted economic definition. □

C.2 Physical projection and the connector/fee region

Proofs of Theorems 11 and 12. With one bus, the state after A is 5 and the state after B is $x - 10$. Early charging has $x \le 10$, so feasibility forces $x = 10$. Terminal replenishment is 20 and requires $K \ge 20$. With two buses, A's bus has terminal demand $15 - x$, B's bus has demand 15, and $0 \le x \le 10$. Serial charging at power K fits the terminal hour exactly when $30 - x \le K$. These conditions also satisfy each battery's individual headroom and inventory bounds. Thus for $K \ge 20$ the two-bus early interval is $[\ell, 10]$, $\ell = 30 - K$; for $K < 20$ neither partition is feasible. At $K = 20$ both projected loads are $(10, 20)$ and their costs differ only by f.

For $K > 20$, strict convexity of g gives the two-bus minimum at $t = \max\{5, \ell\}$. The one-bus minimum is the fixed point 10, proving (12). Projecting complete plans to early energy and operating cost gives the segment $\{(x, 2f) : \ell \le x \le 10\}$ and the point $(10, f)$. Their convex hull has lowest cost

$$c_{\min}(x) = 2f - \frac{f(x - \ell)}{10 - \ell}, \qquad \ell \le x \le 10.$$

The derivative of $c_{\min}(x) + g(x)$ vanishes at $5 + 5f/(2d)$, $d = 10 - \ell$, proving (13).

For $f > 0$, an interior hull minimizer has lower operating cost than every two-bus physical plan at the same early load. Strict convexity excludes the distinct one-bus point from attaining its objective. Therefore the gap is positive exactly when $\ell < h < 10$. Equivalently $\frac{2}{5}d(\ell - 5) < f < 2d$, together with $f > 0$, which gives (14). At an endpoint the physical endpoint attains the hull minimum. For $f = 0$, $h = t$, and the two-bus physical plan attains the hull. Equality of the two physical branch minima gives exactly $f = 90 - g(t)$; within each branch strict convexity fixes its minimizing load. The affine-response formula (15) follows by minimizing over every point of those physical segments.

For $K = 30, f = 7$, the lower hull objective is

$$104 - \frac{27x}{10} + \frac{x^2}{5} = \frac{7591}{80} + \frac{(x - 27/4)^2}{5}.$$

Physical branch values are 97 and 99. The one-bus gradient is $(6, 4)$, and the complete two-bus response at that price is the zero-early endpoint. Bills 147 and 134 give regret 13. The mixture and operating-cost values in the text follow from weight 27/40 on the one-bus point. □

C.3 A reserve/loss neighborhood

Proof of Proposition 13. For reserve $0 \le r \le 5$, constant efficiency $0 < \eta \le 1$, early power P, and terminal power K, define grid-energy quantities

$$T = 30/\eta, \quad \ell = \max\{0, T - K\}, \quad u = \min\{P, 15/\eta\}, \quad h = \max\{\ell, (10 + r)/\eta\}.$$

The two-bus early interval is $[\ell, u]$; the one-bus interval is $[h, u]$ if nonempty. The lower terminal-capacity condition is $T - x \le K$. For the one-bus plan, its state after B is $\eta x - 10$, so reserve imposes $x \ge (10 + r)/\eta$. Charging only A's bus early realizes every interval point; terminal demands fit serially, and the individual bounds follow from the inventory headroom 15 and the stated reserve range.

Write $G_T(x) = F(x, T - x) = G_T(m) + k(x - m)^2$, where $k = 1/5$ and $m = T/2 - 10$. Suppose

$$\ell < m < h < u, \qquad f > 0, \qquad f < 2k(h - \ell)(h - m). \tag{36}$$

On $[\ell, h]$ the hull's lowest operating cost is $2f - f(x - \ell)/(h - \ell)$, obtained from the two-bus endpoint ℓ and one-bus endpoint h; on $[h, u]$ it is f. Put $d = h - \ell$. The hull minimizer $x^* = m + f/(2kd)$

lies in (m, h). Physical and hull values are

$$D = G_T(m) + \min\{f + k(h-m)^2, 2f\},$$
$$\mathrm{CH} = G_T(m) + 2f - \frac{f(m-\ell)}{d} - \frac{f^2}{4kd^2}.$$

Subtracting gives

$$\Delta = \min\left\{k(h-x^*)^2, \frac{f(m-\ell)}{d} + \frac{f^2}{4kd^2}\right\} > 0.$$

At $(r, \eta, P, K, f) = (1, 19/20, 12, 30, 7)$,

$$(\ell, m, h, u, d) = (30/19, 110/19, 220/19, 12, 10).$$

In particular $u - h = 8/19 > 0$ and $2kd(h-m) = 440/19 > 7$. Rational substitution gives the stated values. All inequalities in (36) are strict and the parameter functions are continuous locally. They therefore persist in an open neighborhood of (r, η, P) with $K = 30$ fixed. □

D Expansion and equipment proofs

D.1 The old-plan and dual-certificate statements

Proof of Proposition 14. Set inclusion gives $V_Z(p) = \min\{V_X(p), V_Y(p)\} \le V_X(p)$, so old-plan regret cannot decrease. By Proposition 2, no old plan can become supported. If failure is erased, $D_Z = \mathrm{CH}_Z \le \mathrm{CH}_X < D_X$; thus a new plan attains physical optimum and $D_Y \le \mathrm{CH}_X$. If instead $\min\{D_X, D_Y\} > \mathrm{CH}_X$, then $D_Z > \mathrm{CH}_X \ge \mathrm{CH}_Z$, proving survival.

Under (16), $V_Z(p^*) = V_X(p^*)$, so

$$\mathrm{CH}_Z \ge V_Z(p^*) - F^*(p^*) = \mathrm{CH}_X \ge \mathrm{CH}_Z.$$

For any new physical plan,

$$\begin{aligned} c(y) + F(e(y)) - \mathrm{CH}_X = {} & [c(y) + (p^*)^\top e(y) - V_X(p^*)] \\ & + [F(e(y)) - (p^*)^\top e(y) + F^*(p^*)]. \end{aligned}$$

Both brackets are nonnegative. Equality holds exactly under the fleet and supply equalities stated in the proposition. Compactness ensures that if $D_Z = \mathrm{CH}_X$, some physical plan attains that equality; it cannot be old. With a uniform fleet margin ε, every new physical plan lies at least ε above CH_X, while every old one lies at least Δ_X above it. This gives the quantitative bound. Strict convexity of F makes the minimizer of $F(e) - (p^*)^\top e$ unique on its convex domain whenever attained, proving the projected-point interpretation. □

Proof of Theorem 15. Each added plan buys total energy $30 + w$, $w \ge 0$, because its initial and final battery inventories agree. Its early purchase u is at most 10. For two used buses, the old $(u, 30-u)$ plan of cost 14 is feasible and componentwise no larger in load or cost. For one bus, completion of both services by hour 3 requires $20 + u \ge 30$ even before any additional nonnegative movement consumption. Thus $u = 10$, and the old one-bus plan dominates it. No added feasible plan can reduce the physical objective because F is coordinatewise nondecreasing.

Every hull point is a finite convex combination of physical points. Replace each added component by its old dominator; its mixture has no larger cost or load in any coordinate. Consequently it has no larger convex supply objective. This gives $\mathrm{CH}_X \le \mathrm{CH}_Z$, and inclusion gives the reverse inequality. The same inclusion/dominance argument gives $D_X = D_Z$. This proof does not require the movement timing to be favorable: it concerns every added plan that remains feasible. □

D.2 Battery enlargement and fee thresholds

Proof of Theorem 16 and (19)–(20). The one-new-bus state sequence imposes $x \geq 30 - B = a$ and $x \leq 10$. Every point of $[a, 10]$ is feasible and fully replenished. With two buses, the A bus's headroom after its service is 15 for either type; B's bus starts full before service and cannot buy early energy. Serial terminal charging gives every $x \in [0, 10]$ for every type assignment. Since $q \geq 7$, all mixed/new two-bus projected points are dominated by the old two-bus interval. Therefore

$$D_Z = \min\{97, q + g(t)\}, \qquad t = \max\{5, a\}.$$

Every supported physical plan is a physical planner optimum. No projected old plan can be rescued, and a dominated two-bus addition cannot have zero regret: at equal cost it has an unsupported old counterpart; at larger cost it has a strictly cheaper old counterpart at every price. It suffices to test the new branch's unique physical minimizer t.

If $a \leq 5$, $t = 5$, so $p = (5, 5)$. All plans buy 30 units; comparison with the old one-bus plan is simply $q \leq 7$, hence $q = 7$. If $a > 5$, $t = a$ and $p_1 - p_2 = 2a/5 - 2 > 0$. Within the new branch, the endpoint a minimizes the fixed-price objective. The best old two-bus plan is its zero-early endpoint. Comparison against it gives

$$q \leq 14 - a(p_1 - p_2) = 14 + 2a - \tfrac{2}{5}a^2 = A(a).$$

Comparison against the old one-bus point gives

$$q \leq 7 + (10 - a)(p_1 - p_2) = C(a).$$

These comparisons include all physical alternatives by affine minimization on every branch. They prove (17). For $a > 5$, $C(a) \geq 7$ and $A(a) \geq 7$ exactly when $a \leq (5 + \sqrt{95})/2$ in the relevant domain; also $A(a) - C(a) = 27 - 4a$. At $q = 7$, including $a \leq 5$, this proves (18).

At fee 7 below the battery threshold, the hull's lower boundary joins $(0, 14)$ to $(a, 7)$ and is then flat to 10. The hull objective on the first edge is $14 - 7x/a + g(x)$ and is minimized at $t_h = 5 + 35/(2a) < a$. The physical optimum has early energy a; its gap is $(a - t_h)^2/5$. At and above the threshold the physical point $(\max\{5, a\}, 7)$ attains the hull minimum. Thus the boundary is derived on the whole continuous battery domain, not inferred from a scan.

For $B = 25$, the added endpoint is $(5, q)$. It is below the old lower hull $14 - 0.7x$ exactly when $q < 21/2$. In that regime the new lower boundary joins $(0, 14)$ to $(5, q)$ to $(10, 7)$. The objective decreases along the first edge. On the second it is minimized at $x = 5 + (q - 7)/2$ with operating cost $q - (q-7)^2/10$. Its value is $85 + q - (q-7)^2/20$. At $q = 21/2$ this equals $7591/80$; above it the whole new segment is dominated by the old hull. This proves the formulas. For $q = 8$, substitution gives $\mathrm{CH}_Z = 1859/20$ and $D_Z = 93$. For $q = 12$, both physical minima are 97. At the new load's price $(5, 5)$, the old one-bus bill is cheaper by 5. At the old load's price $(6, 4)$, the old zero-early two-bus bill remains 134; every new one-bus bill is at least $q + 130 \geq 137$. Thus the old plan's regret is 13, including at the fee 12 tie. All quantities use fixed posted prices. □

E Replication proof

Proof of Theorem 17 and Corollary 18. For m paired buses, the complete projected physical set has early energy $E \in [10m, 10n]$, late energy $30n - E$ and operating cost $14n - 7m$. The connector assignment in the text realizes every such early total: paired buses take 10 each, and unpaired A buses distribute the remainder. Convexification has lower operating boundary $14n - 0.7E$. Substitution in (21) gives the hull minimizer $E = 27n/4$ and value $7591n/80$.

For fixed integer $m \geq n/2$, supply is minimized at $E = 10m$ and total cost is

$$104n - 27m + \frac{20m^2}{n} = \frac{7591n}{80} + \frac{20}{n}(m - 27n/40)^2.$$

For $m \leq n/2$, the best early energy is $5n$ and branch cost is $99n - 7m$, decreasing with m. For odd n, moving from $\lfloor n/2 \rfloor$ to $\lceil n/2 \rceil$ changes the minimized cost by $-7 + 5/n < 0$. For even n the branch formulas meet at $n/2$. Thus an integer nearest to $27n/40$ is optimal, including both at a tie, and $|\delta| \leq 1/2$. This proves (22).

At that optimum the own-load marginal price is

$$p = (4 + 2m/n,\ 6 - 2m/n).$$

The optimal m has $m/n \geq 1/2$, so a cheapest two-bus pair response buys zero early (the whole interval ties when $m/n = 1/2$). The one-bus private objective minus that two-bus objective is $40\delta/n$. If $\delta \geq 0$, precisely the m paired choices are disadvantaged; otherwise the $n - m$ unpaired choices are. This proves (23). With one owner, all choices can be replaced simultaneously. To compare separate ownership, first implement the optimum using m fixed-pair owners with one bus and $n - m$ with two. Each one-bus owner buys 10 early; each two-bus owner buys zero early at the planner optimum, giving the same total load and operating cost as the aggregate optimum even if its literal matching crossed the new ownership labels. Reserved connectors make each unilateral replacement feasible. The identical projected opportunity sets give the same complete-fleet response value, while each owner's gain is at most $20/n$. Summing the disadvantaged owners' gains yields the whole-owner expression.

For $n = 40k + 1$, the optimum is $m = 27k + 1$, so $\delta = 13/40$; substitution yields the displayed subsequence. If 40 divides n, $\delta = 0$. At $n = 20$, the tied $m = 13, 14$ have $\delta = -1/2, 1/2$, giving regrets 7 and 14 respectively. All ties concern physical plans; no fractional bus schedule is being implemented. □

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